

Types of innovation

1. Incremental innovation
2. Architectural innovation
3. Modular innovation
4. Radical innovation
5. Disruptive innovation

- Process innovation
- Service innovation
- Open innovation
- Business model innovation
- Social innovation
- Frugal innovation
- Reverse innovation
- Innovation during a crisis

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Most successful innovations are rarely purely incremental, radical, modular, architectural, or disruptive. In fact, firms often combine multiple types of innovation, creating synergies that produce greater customer value and stronger competitive advantage than any single innovation could achieve on its own.



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Scenario: Fast-Food Restaurant Chain

A fast-food chain **continuously improves** its mobile ordering app by making the interface simpler and reducing the number of clicks required to place an order. At the same time, it redesigns its **kitchen workflow** using **automated** order-routing systems so food is prepared faster. The result is shorter waiting times, happier customers, and increased sales.

The above is Incremental and Process innovation.

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Scenario: Video Streaming Platform

A streaming company offers low-cost, on-demand video services that initially appeal to customers underserved by **traditional cable** television providers. Beyond the disruptive business model, it creates innovative services such as personalised recommendations, multi-device access, and ad-free viewing options. The combined innovations **transform** how consumers access entertainment.

The above is Service and Radical innovation.

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Example: Quantum Computing

A technology company creates a revolutionary different quantum computing **architecture** with the potential to solve problems beyond the capabilities of classical computers. Rather than working alone, it **opens** its development platform to researchers, startups, and universities who contribute algorithms, software tools, and applications. All parties help to transform a scientific breakthrough into a commercially viable platform.

The above is architectural and open innovation.

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Scenario: Smartphone Ecosystem

A company reconfigures existing technologies such as cameras, GPS, touchscreens, and internet connectivity into an **integrated smartphone platform**. It then leverages on its ecosystem by allowing **external** software firms to create apps and services. The architectural redesign creates a new platform, while the apps and services created by the collaborators dramatically expand their value to users.

The above is architectural and open innovation.
or modular

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Scenario: Automobile Manufacturer

A car manufacturer introduces **small improvements** to fuel efficiency and driver-assistance features in each new vehicle model. Simultaneously, it adopts robotic quality-inspection systems that reduce production errors and manufacturing costs. Neither innovation is revolutionary on its own, but together they strengthen competitiveness and profitability.

The above is incremental and process innovation.

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completely different way
people may feel not comfortable

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radical

Scenario: mRNA Vaccines

A biotechnology firm develops a **distinctly new** mRNA vaccine platform that differs fundamentally from traditional vaccine technologies. To accelerate development, it collaborates **openly** with universities, government laboratories, and pharmaceutical partners around the world. The breakthrough technology combined with external knowledge and resources enables rapid commercialisation and global impact.

The above is radical and open innovation.

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Scenario: Telemedicine Provider

A revolutionary telemedicine platform enables patients to consult doctors remotely, initially serving people who find traditional healthcare inconvenient or inaccessible. The platform also supports online prescription management, digital health records, and 24-hour virtual consultations. The result is a **more accessible and customer-centric** healthcare experience.

The above is disruptive and service innovation.
or radical

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Scenario: Smart Farming Solution

An agricultural technology firm combines existing sensors, drones, weather data, and AI analytics into a new integrated farming system. To improve the solution, it collaborates with farmers, agricultural researchers, and equipment manufacturers who contribute ideas and specialised expertise. Success comes not from inventing entirely new technologies, but from rearranging existing components and leveraging external knowledge.

The above is radical and architectural innovation.

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